Artist Statement, Interactive Bike Garage

By Émile Bédard 40343664

Work presented to Sabine Rosenberg

CART 263,Creative Computation II, Section A

Computation Arts, Concordia University ─ March 8th, 2026

URL of the project (desktop and mobile):
https://emilebedard.github.io/cart263/Projects/project1/Interactive_Bike_Garage/

Interactive Bike Garage is made for cycling enthusiasts to register bike parts, maintain their vehicles and enjoy their passion of cycling. It serves as a comprehensive inventory for bike parts and a record for both daily upkeep and professional shop visits.

The experience was built in response to a simple problem: how can amateur cyclists learn parts names and gain mechanical literacy. The cycling community is still perceived as marginal in North America and it can be daunting to start the hobby. Many new cyclists don't know the basics before riding a bike and can suffer from this lack of knowledge when they encounter mechanical failures or when their bike needs general maintenance. Interactive Bike Garage serves this purpose as a proof of concept, a working prototype to accompany new enthusiasts in this intimidating world. Consequently, the web interactive experience is standing for much more, it is restating the power of databases as empowering actions and it sprouts in a political and cultural "bikelash" that we are presently experiencing in Montreal. I will develop much more on these two fundamental topics after addressing the core features of the web application and its development process.

Interactive Bike Garage (IBG) is the final form of a few interlinked HTML5, CSS and JavaScript files hosted online. It interacts with a user through the medium of a web browser using inbuilt features of this medium such as event listeners and the browser's local storage. Therefore, the project was developed around this last feature due to its longevity aspect, allowing the user to reliably use the tool. IBG consists of three HTML pages: one main landing page, one registration page for parts names and one final page for maintenance quick tips. The main part of the work is concerning the second page: "register parts". It presents the user with a simple and stylized interface featuring buttons and a form that appears after a certain action. The user is prompted with one call to action at the top of the page: "Add a new bike or select one to register parts". This unique instruction then leads visitors to add a bike using the similarly named button, registering a bike name, type and color. Then the created bike appears and can be selected to unveil the form and register all of its most known and discussed components such as derailleurs, wheels and groupsets. The form appears side by side with a personalized bicycle infographic showing where the said parts are located on a general bicycle. All of this data is then submitted and saved in the user's browser local storage to reappear the next time they visit the website. The third page, maintenance, is mainly an HTML page with little interactions. It serves the second purpose of the experience: democratizing mechanical literacy and allowing users to autonomously learn their bike inner workings and gain more mechanical and repairability freedom. It takes shape as a list of quick tips ranging from super-beginner level to more essential reminders for established bike repair enthusiasts.

The initial idea for the project was to recreate a tool I made myself in Notion using their automation functions. The project then started to diverge and concentrate on bike parts collection and inventory and I got influenced with my visits at local coop bike shops and my conversations with fellow cyclists. I realized that many of us don't know basic stuff and that it is very difficult to gain this knowledge autonomously. Most of the time, we are taught by others around us. This then leads to another major problem involving peer to peer teaching in this field: mansplaining and the problem of women representation in bike shops and the cycling community. I won't address this topic in my statement because it concerns a much larger debate but it is a core problem of this milieu that I wanted to underline. The most difficult part of the project was figuring out the database logic and how to store this information in the local storage. The CSS styling was also a nightmare to figure out. I coded the project manually on VS code but encountered many issues where I needed to look at solutions online or prompt GenAI for solution propositions. I used some generated code in my project and most of it was manually reworked after, generally what was generated is pointed out with inline comments.

I decided to adopt a data driven project because I was compelled by the value and satisfaction of database organization. Data can be beautiful and how we display it can radically change our perception of it; this is something marketers are well aware of and never doubt to capitalize on but data collection is also used by artists like Georgia Lupi and Stefanie Posavec with their analog drawing data piece called "Dear Data"[1]. Their work is a collection of memories, data representing their days and weeks revolving around a weekly theme. This project inspires me a lot to see data as something more than numbers, but meaningful memories. With this in mind, data for me is a very interesting medium, it can take many shape: keywords, images, elements and much more but the form I will be focusing on is a JSON type of database: a collection of text values associated to labels, variables or keys, all meaning a similar relationship here of label-value. Collecting databases is empowering, but how ? Because registration and retrieval of information is something familiar and natural to us humans, it is intuitive, we do it countless times everyday with our brains and our memory. When we relate this aspect of the human condition to the nature and shape of numerical databases it is only logical to understand how and why we built these tools this way: we already knew something similar. Recollection is building positionality and embedding deep roots to grow and bloom identity. We are a synthesis of our

[1] Lupi, Georgia and Posavec, Stefanie.2016. Dear Data. https://www.dear-data.com/theproject

memories and experiences, we learn, we live following these conscious and unconscious guidelines to exist. But with no experience and no memories we can't deduct answers and create. And this is how data is empowering, because it creates foundational pillars for actionable decisions. Manovich states in his text *Database as a Genre of New Media:* "database and narrative are natural enemies. Competing for the same territory of human culture, each claims an exclusive right to make meaning out of the world."[2] Here, he defines the line that separates databases and narratives and proves a point: they try to make meaning out of the world, just like we try to rationalize our environments with memories and we gain power with data.

Over data and parts names, IBG is also standing as an advocacy for the "Masse Critique" movement happening in Montreal and the cultural "bikelash". Let's start by defining this term: it is a term to define the vivid opposition facing cyclists after the installation of a new cycling infrastructure like a protected bike path. Montreal is torn between two ideologies here: opposing the traditional north american city where cars are king to the european model that favor active transport and decenter the car street as a city's foundation. Like the youtube channel "Not Just Bikes"[3] noted in its video about the Montreal bike infrastructures, it is a city fractured by car-centric, wide-street history and political and ideological evolution and need for change. This debate is very active and with the recent municipal election, bicycles became the central symbol of discourse and people started to categorize and label citizens based on their transportation methods. What's interesting too is that being located in North America, Montreal experiences strong winters with heavy snowfall but it still manages to stay a cycling city! Urbania addressed this topic in their recent February micromag titled: "Vélo Sous Zéro". In the coverage, one interviewee states: "Even though bike paths only represents only 10% of our infrastructures, that's what drives attention and strike controversy"[4]. Here we are clearly talking about the radical bike changes announced by the new party at the head of the city elected last November. Ensemble Montréal embodies this "bikelash" in the political discourse in response to the many bicycle and mobility innovations brought by Projet Montreal with its last mandate.

Interactive Bike Garage is more than just an integrated web tool, it is a form of user-centric data visualization manifest to manually recollect and empower cyclists in urban context where choosing to adopt a cycling habit is a politicized action.

---

[2] Manovich, Lev. 2000. Database as a Genre of New Media. https://link.springer.com/article/10.1007/BF01205448
[3] Not Just Bikes. 2023. I Visited the Best* City in North America (Montréal). https://youtu.be/_yDtLv-7xZ4?si=eqmkK0R9cuJbsWuK
[4] Bourbeau, Jean. Urbania 2026. Vélo Sous Zéro. https://urbania.ca/micromag/velo-hiver-montreal-micromag-144